

IMP QUESTIONS

1) A relight is :

- A) The initiation of the after burning system.
- B) What occurs when the engine drain valve is stuck.
- C) The action of the restarting a flamed out engine usually while airborne.**

2) Choose the correct statement:

- 1. AFT CG limit is set to ensure a minimum degree static longitudinal stability
- 2. FWD CG limit is set to ensure a minimum degree of controllability under the worst circumstances

- A) 1 is correct
- B) Both 1 & 2 are incorrect
- C) Both 1 & 2 are correct**

3) Swept back and Delta wings are:

- A) Very efficient at high speed but corresponding inefficient at low speeds**
- B) Fairly inefficient at high speed but very efficient at low speed
- C) Neither efficient at low or high speed

4) The function of Shut off valve in a pneumatic system is to:

- A) Shut Off the pneumatic system
- B) To switch off bleed air supply from an engine and the APU**
- C) To regulate the supply of bleed air

5) Shock stall is caused is due to:

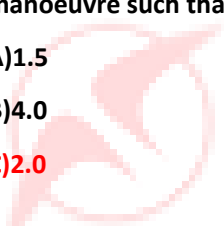
- A) Abrupt movement of the control column
- B) Sudden increase in lift
- C) Boundary layer separation aft of the shockwave causing sudden loss of lift**

6) The main contributory factors which cause creep in turbine blades are:

- A) High RPM and High gas speed.
- B) High Temperature and High gas speeds.
- C) High Temperature and Tensile loading**

7) In level flight, since lift equals weight, if the aircraft is performing a manoeuvre such that, twice the weight, the load factor will be:

- A) 1.5
- B) 4.0
- C) 2.0**

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FLIGHT TRAINING

8) In turbo-compressor the cooling air is:

- A) Ram air**
- B) Compressor air
- C) Cabin air

9) Nucleus has a:

- A) Positive charge**
- B) Negative charge
- C) Positive & Negative

10) Mark the correct statement for climb performance?

- A) Higher the mass, poor climb performance
- B) Speed of the A/C for higher ROC is increased with increased mass
- C) All above are correct**

11) A complete breakdown of airflow through a compressor is known as:

- A) Compressor turbulence
- B) Compressor Buffet
- C) Compressor Surge**

12) Positive static stability is:

- A) If an A/C has a tendency to continue in the direction of disturbance.
- B) If an A/C is subject to disturbance and has neither the tendency to return nor the in the displacement direction.
- C) If an A/C is disturbed from equilibrium and has the tendency to return to equilibrium**

13) What is phugoid oscillation?

- A) Short period in pitch attitude, airspeed, altitude and AOA vary
- B) Long period oscillation in pitch attitude, airspeed, altitude remain constant but AOA vary
- C) Long period of oscillation in pitch attitude, Speed attitude vary but AOA remains relatively constant**

14) The purpose of the diffuser vanes in a centrifugal compressor is to:

- A) Increase the charge temperature
- B) Convert Kinetic energy into pressure Energy**
- C) Convert pressure energy into kinetic energy

15) Increasing supply frequency with constant(regulated) voltage will cause the amplitude of the current wave to----

- A) Inc
- B) Reduce
- C) Remain same**

16) As per Bernoulli's theorem –

- A) Decrease in speed along a stream line point decreases static pressure
- B) Increase in speed at a point along stream line, Cause a decrease in static pressure at the point**
- C) Increase or Decrease in speed has no change in static pressure

17) What is most fav CG location for fuel economy?

- A) CG located at forward CG limit
- B) CG location be same as CP location
- C) CG located at the aft CG limit**

18) Thermal efficiency of engine is increased by :

- A) Increasing the combustion chamber volume
- B) Increasing the R.P.M
- C) Increasing the compression ratio**

19) A typical APU can provide

- A. Air for engine starting
- B. Electrical power for ground or inflight use
- C. All of the above**

20) Load factor is:

- A) Weight divided by lift expressed as 'g'
- B) CL divided by weight
- C) Lift divided by weight expressed as 'g'**

21) Ionisation smoke detector work on the principle of:

- A) Measurement of light transmissibility by a photo electric cell
- B) Visually illuminating of smoke samples
- C) Through the process of ionisation that takes place continuously inside a detection chamber for presences of smoke particles.**

22) What is the purpose of a hydraulic check valve?

- A) To direct the fluid in various directions
- B) To restricts the fluid flow in one direction
- C) To obtain fluid flow in one direction**

23) Control "overbalance" results in :

- A) A gradual increase in stick force with nose down tendency
- B) A gradual loss of control followed by nose up tendency
- C) A sudden reduction then reversal of stick force.**

24) The power increase that occurs with initial increase in altitude when an engine has supercharger fitted is due to:

- A) The increasing charge temperature
- B) The reducing exhausts back pressure**
- C) The decreasing density of the atmosphere

25) The rate at which work is done called:

- A) Current
- B) Power**
- C) Resistance

26) Because of the reduction in density of the atmosphere, with an increase in altitude:

- A) The octane rating of the fuel must be increased
- B) The throttle must be reduced to maintain the best Air/Fuel ratio
- C) The mixture control must be moved towards the weak position**

27) Dynamic stability is ----- at high altitude due to ----- aerodynamic damping

- A) Increased Reduced
- B) Reduced Reduced**
- C) Reduced Increased

28) Why do wind shields consist of 5 layers?

- A) To prevent fatigue
- B) To enable effective heating
- C) To increase their impact resistance**

29) Electrical resistance is measured in:

- A) Ohm**
- B) Watt
- C) Amp

30) Rudder gives :

- A) Lateral control about the longitudinal axis
- B) Longitudinal control about the lateral axis
- C) Directional control about the normal axis**

31) Negative static stability:

- A) Exists if an A/C has tendency to continue in the direction of disturbance**
- B) Exists if an A/C is subject to a disturbance and has neither the tendency to return nor the tendency to continue in the displacement of direction
- C) Exists if an A/C disturbed from equilibrium

32) The current in a circuit can be adjusted by:

- A) reducing the current
- B) Using different materials
- C) Altering the resistance**

33) What is the purpose of a hydraulic reservoir?

- A) To convert hydraulic pressure into work
- B) To store hydraulic fluid**
- C) To support the system with pressurised fluid

34) Materials which have free electrons are called:

- A) Insulators
- B) Resistance
- C) Conductors**

35) A choked nozzle;

A) Gives additional pressure without the addition of fuel

B) Has no effect on thrust

C) Implies that no further increase in velocity can be obtained without the increase of heat

36) Choose the correct statement

A) High performance sailplanes have high AR

B) jet fighter aircraft have low aspect ratio

C) Both A & B

37) What is the effect of ice, frost and snow on stall?

A) Change aerofoil section increases stalling angle decrease in stall speed

B) Change aerofoil section decrease stall angle and increase in stall speed

C) Change aerofoil section increase stall angle increase in stall speed

38) The A/C is at best L/D ratio when parasite drag is ;

A) At minimum

B) Equal to induced drag

C) Greater than induced drag

39) Longerons are beams in the fuselage which are fitted:

A) Longitudinally

B) Horizontally

C) Diagonally

40) Choose the correct statement:

- A) Decreasing the camber of an aerofoil will increase its lift
- B) Increasing the camber of an aerofoil will increase its lift**
- C) Increasing the camber of an aerofoil will decrease its lift

41) Delta-wing airplanes have moveable control surfaces on the trailing edge. They are called:

- A) All moving tail
- B) Stabilizers
- C) Elevons**

42) Power is measured in units of:

- A) Ampere
- B) Watts**
- C) Voltage

43) Choose the correct answer for MONOCOQUE construction:

- A) All the loads are taken by the skin**
- B) This type of construction is good for heavy aircraft
- C) Loads are shared by the skin and the structure

44) A compressor stall causes:

- A) The vibration level to increase with a decrease in the turbine gas temperature
- B) An increase in the turbine gas temperature and the vibration level**
- C) The rotation of the engine to stop suddenly

45) An atom has gained an electron is called a:

- A) Static 10N
- B) Neutral 10N
- C) Negative 10N**

46) Choose the correct statement about aerodynamic heating:

- A) It is not very significant up to speed of mach 2.0
- B) It shortens the fatigue life of the skin
- C) Both of the above**

47) Mark the correct statement:

- A) Power effect most prominent when aircraft has high power low speed during take off and while approaching**
- B) Instability depends on distance from CG to propeller and is lowest at high power and low dynamic pressure
- C) Combined direct & indirect power effects increase of static stability at high power and high lift

48) What are the factor affecting stall?

- A) Mass of a/c, load factor, use of power, CG, Lift coefficient**
- B) Mass of a/c, use of high power, higher power, CP, drag coefficient
- C) Mass of a/c, load factor use of power, lift and drag

49) The power to start an APU comes from:

- A) Ground power unit
- B) Aircraft main DC battery**
- C) Aircraft main engine generator

50) When connecting battery in series the total voltage:

- A) Sum of the battery
- B) Equal to the individual battery voltage
- C) The sum of their individual voltage**

51) An increase in air density will ----- engine power and ----- propeller drag at a particular throttle and propeller setting:

- A) Increase Decrease
- B) Increase Increase**
- C) Decrease Increase

52) A restrictor valve –

- A) is used to restrict the number of services available after loss of system pressure
- B) Controls the rate of movement of a service in one direction**
- C) Controls the rate of build-up of pressure in the system

53) Centre of pressure moves:

- A) FWD at increased transonic speed
- B) Remains constant in a sweep back wing
- C) Aft at increased transonic speeds**

54) Dry Chemical (Dry powder):

- A) Stored in purple, brown, or green containers and is very effective against electrical and flammable fires
- B) The use of this agent in crew compartment or passenger cabins of pressurized aircraft is not permitted. It is best known for its applicability against wheel and brake fire**

C) It is principal extinguisher for use on flammable liquid fires. It blankets the flame by excluding oxygen

55) How do large aircraft equipped with tricycle landing gear obtain steering control on ground?

A) By use of nose wheel steering

B) By use of rudder

C) By use of differential braking

56)) When an A/c structure is designed according to fail safe philosophy it means

A. The structure will never break regardless of stress applied

B. The failed structure can be replaced easily

C. Adjacent structure with stands the loads which the failed structure should have carried

57) About which axis roll is achieved?

A) The horizontal axis

B) The vertical axis

C) The longitudinal axis

58) Control locks are used to prevent:

A) Controls from wear and tear

B) Unauthorised interference with them

C) Mechanical damage in strong and gusty winds

59) The dihedral construction of an air craft wing causes:

A) A directional stability about the lateral axis

B) Lateral stability about the longitudinal axis

C) Longitudinal stability about the lateral axis

60) The control which gives lateral control is:

A) Elevator

B) Rudder for control in yaw about normal axis

C) Ailerons for control in roll about longitudinal axis

61) Lift and drag on airplane depends upon?

A) C.P, C.G, IAS , Air density

B) AOA, air density, velocity, CG

C) AOA, air density, aerofoil shape , velocity

62) The nozzle is said to be "choked" when:

A) The gas flow through it is subsonic

B) The gas temperature rises

C) The gas flow through it reaches its sonic value

63) Choose the correct statement, Taper ratio is:

A) Ratio of average chord to root chord

B) Ratio of root chord to tip chord

C) Ratio of tip chord to root chord

64) An indication of the battery state of charge or serviceability is:

A) The specific gravity of the electrolyte

B) The temperature of electrolyte

C) To check the temperature of electrolyte using a hydrometer

65) Electro Motive Force(EMF) is a force behind the current to make it flow.

If the EMF:

A) Decrease, the flow of electrons increases

B) Increase, the flow of electrons decreases

C) **Decreases, the flow of electrons decreases**

66) Fire detection system generally use four methods detection :

A) FWD cargo, AFT cargo, APU and engines deflection system

B) AFT cargo fire bottles APU fire bottles engine fire bottle & FWD cargo fire bottle

C) **Melting link detectors, Differential expansion detectors, Gas filled detectors and continuous fire detection**

67) If landing is made without flaps the landing speed must be:

A) reduced

B) **Increased**

C) No change in speed is required

68) What is V_{so} ?

A) **V_{so} stalling speed with flaps down and landing gear down.**

B) V_{so} stalling speed with flaps up

C) V_{so} stalling speed with flaps fully down

69) With increase in OAT, SFC will?

- A) Decrease
- B) Increase**
- C) Stays the same

70) What is the purpose of a hydraulic sequence valve?

- A) To activate the shuttle valve
- B) To sequence certain events**
- C) To activate fire shut off valve

71) How are wing bending moments reduced in flight?

- A) by having wing mounted engines and using the wing fuel first
- B) By using ailerons up float and using centre tank fuel as late as possible
- C) By using aileron "up float" and using wing fuel as late as possible**

72) When air is pressurised by an engine driven compressors, it is also :

- A) Cooled
- B) Heated**
- C) Moisturized

73) "MACH TRIM" or "MACH UNDER" are associated with:

- A) Flying at low speed and high gross weight
- B) Flying at supersonic speeds**
- C) Flying at high subsonic speeds

74) EMF is measured in :

- A) Units of Voltage**

B) Units of Amp

C) Units of watt

75) When responding to a TCAS 2 resolution advisory, the crew must respond within:

A) 2.5 Seconds using g forces $\pm 0.25g$

B) 5 Seconds using g forces of $\pm 0.25g$

C) 5 Seconds using g-forces of $\pm 0.5g$

76) The on load/nominal voltage of each cell of a lead acid battery is -----and on load voltage of one cell of Ni-Cad battery is:

A) 1.27 Volts / 1.24 Volts

B) 2.0 Volts / 1.27 Volts

C) 2.0 Volts / 1.2 volts

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77) Icing on aircraft:

A) can change the stall characteristics and stall warning device may not function

B) Can contaminate wing and tail surfaces, engine and pitot- static system

C) All of the above are correct

78) The toilet fire detection must consist of:

A) Fire bottles, Fusible plug and spray ring in addition with smoke detector system

B) Water glycol detection system

C) fire detection system

79) An aircraft in level flight and cabin altitude increases pressure differential will :

- A) Remain the same
- B) Increase
- C) Decrease

80) What is the solidity of propeller?

- A) Ratio of total frontal area of the blades to the area of propeller disc
- B) Ratio of total frontal area angle to L/D ratio
- C) Increasing number of blades increases aspect ratio

81) The altitude above which icing does not occur since the droplets are all frozen and will not adhere to the aircraft's surface is:

- A) 37,000 ft
- B) 42,000 ft
- C) 40,000 ft

82) The aeroplane control surface which gives directional control about normal axis is?

- A) Rudder
- B) Ailerons
- C) Elevators

83) When connecting batteries (of the same type) in parallel the total voltage is equal to:

- A) the sum of their individual voltage
- B) The individual voltage battery voltage

C) Sum of the battery current

84) What is high speed buffet and its effect on aircraft?

A) It can occur at any altitude with maximum operating speed

B) It can occur at high altitude with little effect on airframe structure

C) It is the same as a shock stall, Can seriously damage aircraft structure at high altitude.

85) The highest pressure in the gas turbine engine occurs:

A) between the compressor and combustion chamber

B) In the combustion chamber

C) In the jet pipe

86) The maximum thrust that a jet engine can develop will be:

A) Go around thrust

B) Max Climb thrust

C) Max static thrust

87) Chemical oxygen generators:

A) The generators are relatively light self-contained devices.

B) Sufficient oxygen is supplied for a period of minimum 30 minutes.

C) A relief valve prevents internal pressure in the generators exceeding 50 psi as the normal flow psi

88) The purpose of the tertiary airflow in the combustion chamber is to:

A) Ensure complete combustion of the fuel

B) Ensure the gas temperature and cool the burner head

C) Reduce the gas temperature and cool the flame tube

89) The disadvantage of "Conventional" elevator and trim tab is that aircraft?

- A) nose up authority reduces with forward CG movement**
- B) Nose up pitch authority increase with FWD CG movement
- C) There is no effect of CG movement to pitch authority

90) Ventral fin may be added as in additional contribution to -----

- A. Directional stability**
- B. Lateral stability
- C. Static lateral stability

91) How do we achieve effective angle of attack over the propeller blade ?

- A) By increasing blade angle from propeller root to propeller tip
- B) Reducing blade angle from propeller root to propeller tip**
- C) Increasing Blade angle from propeller tip to Propeller Root

92) How is the ice prevented from forming on the front wind shield of large system:

- A) Warm air is blown over the wind shield
- B) De-icing fluids are sprayed
- C) Electrically heated wind shields**

93) Dynamic hydroplaning is recognised by:

- A) Total loss of control
- B) The tyre skids all the time
- C) Partial loss of controls**

94) A propeller which is wind milling:

- A) Rotates the engine in reverse and gives drag
- B) Rotates the engine in reverse and gives some thrust
- C) Rotates the engine in the normal direction and give drag**

95) When gases passes through convergent duct:

- A) Vel and temp Inc and their pressure decreases
- B) Vel inc and temp and pressure dec**
- C) Vel dec and temp and pressure inc

96) In the event of an engine surging occurring, the correct action to be taken is :

- A) Close throttle quickly
- B) Close throttle slowly**
- C) To open throttle fully

97) With asymmetric thrust mark the correct statement regarding effect of airspeed?

- A) Effectiveness of controls depends upon static pressure
- B) Lower IAS will reduce the available rudder moment**
- C) Both A and B are correct

98) Ohm's law can be summed as :

- A) Current inversely proportional to voltage if voltage remains constant
- B) Current directly proportional voltage:
- C) Current inversely proportional to resistance if voltage remains constant**

99) Current is :

- A) A magnetic field produce around the conductor
- B) Potential difference across the conductor
- C) The number of electrons passing any point in the conductor in one second**

100) A fuel sample appears cloudy or hazy could be due to;

- A) Dirt in the fuel
- B) Oil in the fuel:
- C) Water contamination**

101) Stress is defined as :

Internal force inside a load

Force per unit area measured in MN/m²

Twisting force inside a structural member

102) A fuel strainer should be fitted:

- a. in the inlet manifold
- b. at the air intake
- c. before the main jet**
- d. after the main jet

103) The bypass ration of the engine is the ratio of :

Primary air to tertiary air

Cold stream air to that flowing through the hot core of the engine

Exhaust gas pressure to air intake pressure.

104) Bleed valves are automatically opened:

At maximum RPM to prevent compressor stall (low RPM)

At low RPM to prevent the turbine stalling

During engine acceleration to prevent turbine surge.

105) The pressure within the gas turbine engine with it is being run on a stationary A/c is-----ambient pressure

- A. Above
- B. Same as
- C. Below**

106) The interstage airseal is used when

- A. Engine sections are operating at different pressure**
- B. Engine sections are subjected to the same value
- C. It is more convenient

107) The term (self sustaining speed) means that

- a. the aircraft can roll forward with no further opening of the throttles
- b. the speed from which the engine can accelerate to full power within 5 seconds
- c. the engine will run independently of external help
- d. the speed from which the engine can accelerate to idle without the help of the starter motor**

108) In a turbo fan engine, the fan speed is controlled by

- A. A reduction gear
- B. A waste gate
- C. The turbine**

109) A typical APU can provide

- A. Air for engine starting
- B. Electrical power for ground or inflight use
- C. All of the above**

110) A cannular combustion system

- A. A set of flame tube each of which is mounted in a separate air casing
- B. A set of flame tubes enclosed in a common air casing**
- C. One common flame tube enclosed in a common air casing

111) The primary function of a diffuser in a carburettor is to:

- a. control the mixture strength over part of the engine speed range**
- b. vent air from the float chamber
- c. emulsify the fuel during engine acceleration
- d. enable adjustment of the engine slow running speed

112) The current in a circuit can be adjusted by:

Resistor

Inverter

Rectifier

113) Fuel booster pump are fitted:

Fitted inside the tanks

Fitted outside the tank

Both A & B correct

114) A hydraulic gear extension/retraction mechanism consists of sequence valves, uplocks and:

- a. an anti-skid braking system
- b. **downlocks**
- c. torque links
- d. a shock absorber

115) What is viscosity?

Resistance to flowing liquid

Resistance to current

Both of the above

116) Electrolytes :

Encourages electron transfer between the electrodes until there is a potential difference between them

can be used to store charge and electric current

Both A & B

117)) Lead battery acid battery are stored in a

A. Semi charged Charged state to prevent deuteriation of the battery by sulphation

B. Discharge state and have a longer shell life

C. Charged state

118) Factor affecting the resistance

A. Type of material

B. Longer wire will have less resistance

C. Thicker the wire higher the resistance

119) Class D compartment (which can not be reached by a crew inflight)

A. Cabin crew operated fire extinguisher

B. Must not be ventilated to contain the fire

C. The compartment shall be lined with fire resistant material and ventilation and draughts must be there so that any fire likely to occur will not progress beyond safe limits

120) Major advantage of Semi Monocoque structure over a monocoque structure is:

A. Better strength to weight ratio

B. Cheaper to manufacture

C. It is lighter

121) Battery capacity percentage efficiency must always be:

a. 10% above saturation level

b. above 70%

c. 80% or more

d. above 90%

122)) Shuttle valves will automatically:

Switch hydraulically operated units to the most appropriate pressure supply.

Shut down systems which are overloaded.

Guard systems against overpressure.

Reduce pump loads.

123) The purpose of a ditching control is to:

achieve rapid depressurisation.

open the outflow valve(s).

direct pressurisation air to the flotation bags.

close the outflow valve(s).

124) The ""cabin differential pressure"" is:

approximately 5 psi at maximum.

cabin pressure minus ambient pressure.

the pressure differential between the air entering and leaving the cabin.

125) Cabin pressure is controlled by :

delivering a substantially constant flow of air into the cabin and controlling the outflow.

controlling the flow of air into the cabin with a constant outflow.

the cabin air re-circulation system.

the cabin air mass flow control inlet valve(s).

126) The ice protection for propellers of modern turboprop aeroplanes works pneumatically.

with hot air.

with anti-icing fluid.

electrically.

127) The fuel system boost pumps are submerged in the fuel.

To prime the pumps.

Because their efficiency is greater.

To shorten the fuel lines, so minimising the pressure losses.

To cool the pumps.

128) A static inverter is a:

device for reversing the polarity of the static charge.

static discharger.

transistorized unit used to convert DC into AC.

129) The most widely used electrical frequency in aircraft is :

20 Hz.

400 Hz

200 Hz

130) A thermal circuit breaker:

protects the system in the event of overheating, even without exceeding the maximum permissible current.

is a protection system with a quick break capacity of about one hundredth of a second.

forbids any overcurrent.

can be reset without any danger even if the fault remains

131) Fire extinguisher to be use on wheel and brake fire:

Dry powder(Chemical)

Water

CO2

Foam

132) If resistance increases with an increase in temperature, the resistor is said to have:

Positive Temperature Coefficient (PTC)

Negative Temperature Coefficient (NTC)

None of the above

133) Most ideal shape cross sectional shape for pressurised fuselage is:

Oval

Double bubble

Circular

134) Longerons are:

Run lengthwise along the fuselage joining the frames together
 Responsible to provide space for electrical cables
 Both of the above

135) 3 Basic type of fuses

Cartridge, HRC & Trip free
Cartridge, HRC & Current limiters
 Trip free & non-trip free

136) In order to reduce adverse aileron yaw A/c are fitted with

- A. Servo balance yaw
- B. Differential ailerons**
- C. Differential balance yaw stick force reduce to 0.

137) In modern aircraft L/D ratio is 10 and 20 to 1 which means

- A. Lift is 20 times drag is 10 times
- B. Lift is 10 times drag is 3 times
- C. Lift is 10 to 20 times greater than drag**

138) Advantages of the Fly by wire (FBW) system over conventional flight control system are:

- A. It is lighter and does not require the heavy control cables
- B. Low weight leads to better fuel consumption
- C. Both of the above**

139) How are the forces are balanced in a steady power on descent?

- A. Weight acts as two components: One component balance lift and the second one acts forward to add thrust

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B. If the nose of the aircraft is lowered with a constant thrust setting, the increased component of weights acting forward will increase IAS

C. **Both A and B are correct**

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140) *What is incipient stall?*

A. *Wingroot is stalled but not completely*

b. *Leading edge is stalled but not completely*

c. ***Wing has begun to stall in few areas but not completely***

141) *The airflow behind an oblique shockwave will*

a. *Always be subsonic*

b. *Could be subsonic or supersonic*

c. ***Always supersonic***

142) *When does the bow wave first appear?*

a. *At MCRIT.*

b. *At Mach 1.*

c. ***Just above Mach 1.***

d. *Just below Mach 1.*

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143) An engine failure during Take off of a twin engine aircraft results in:

High loads on vertical stabilizer

High load on the main wheel

High loads on horizontal stabiliser

144) Choose the correct answer for reducing induced drag

A. Having winglets

B. Shaping the wing tips

C. **Both A and B**

145) Choose the correct statement

- A. High lift devices are fitted on leading edge only
- B. High lift devices are fitted on trailing edge only
- C. Can be fitted on the leading and trailing edges**

146) Operational rough air speed VRA/MRA should consist of the following feature

- A. Be maximum structural cruise speed is with ability to with gust loads
- B. For utility category aircraft the positive load limit above is 4.4 and 1.76 the negative limit load factor
- C. Should be high enough to avoid staling and low enough to avoid damage to the structure**

147) The angle between blade chord and the plane of rotation is

- A. Blade angle**
- B. Geometric pitch
- C. Angle of attack

148) A Bypass Ratio of 5:1 means that:

- a. 5 pounds of air is bypassed for every 10 pounds entering the engine intake
- b. 5 pounds of goes through the HP compressor for every 10 pounds that enters the intake
- c. 10 pounds of air goes through the bypass for every 5 pounds that enters the intake
- d. 5 pounds of air is bypassed for every 1 pound that goes through the hot core of the engine**

149) One stage of an axial flow compressor consists of:

- a. one rotor assembly and one row of stator vanes**
- b. one stator assembly and one row of guide vanes

- c. one rotor and one impeller assembly
- d. one impeller and one diffuser assembly

150) The extra space in the oil tank is to cater for:

- a. frothing and aeration of the oil as it passes through the engine.
- b. fire protection.
- c. the accommodation of extra oil contents on long duration flights.
- d. anti-surge action.

151) Lox is:

- a. Lean Oxygen
- b. Low pressure oxygen pressure
- c. Liquid oxygen

152) What is the percentage increase in stalling speed in straight and level turn?

- a. 30 degree angle of bank 9% Increase in stalling speed
- b. 45 degree angle of bank 21% Increase in stalling speed
- c. 70 degree angle of bank 100% Increase in stalling speed

153) Lead acid batteries are stored in a:

- a. Semi charged state to prevent deterioration of the battery by sulphation
- b. Discharged state and have a longer shelf life
- c. Charged state

154) Batteries which are susceptible to thermal runaway:

- a. Ni-cad

- b. Lead acid
- c. Both a & b

155) A NiCad battery shows a high temperature after engine start, this could be an indication of:

- a. **thermal runaway**
- b. it is not connected to the battery bus bar
- c. normal temperature during charging
- d. depends upon the outside air temperature

156) When throttles are advanced for take off the configuration warning will sound if:

- a. The trim position is outside the take off range
- b. The speed brakes are not in the selected position
- c. **Any or all of the above**

157) The influence of ground effect is experienced by an aircraft when its height above ground is:

- a. About one wing span
- b. **About half wing span**
- c. About one and a half wing span

158) Fire detection system must be capable of :

- a. Automatically operate the fire extinguisher system
- b. Automatically after two minutes or aural warning operate the fire extinguisher system
- c. **Provide rapid detection of a localised fire or overheat conditions however it must not automatically operate the fire extinguishers.**

159) What is the function of flap load relief system?

a. Pilot can select the flap but will not extend until speed is below flap extend speed

b. Flaps will extend as soon as speed decreases to its value

c. Both A & B are correct

160) The purpose of an air inlet is to provide a relatively supply of air to the of the compressor.

a. turbulent free face low pressure

b. turbulent face low pressure

c. turbulent free rear low pressure

d. turbulent free face high pressure

161) A typical APU can provide:

a. Air for the engine starting

b. Electrical power for ground or in flight use

c. All of the above

162) The phenomenon of flutter is described as:

a. Reversal of the ailerons caused by wing torsional flexibility

b. Rapid movement of the airframe caused by the vibration from the engines

c. Oscillatory motion of parts or parts of the aircraft relative to the remainder of the structure.

163) The critical Mach number of the whole aircraft is the speed at which any disturbance may affect the :

a. Controllability of the aircraft

b. Lift or drag of the aircraft

c. Both of the above are correct

164) Secondary cells can:

- a. Not be recharged
- b. Can be discharged and recharged many times over a long period of time**
- c. Can be recharged only once

165) EMF is measured in units of :

- a. Voltage**
- b. Potential difference
- c. Both (b) & (c) are correct

166) When an aircraft structure is designed according to fail safe philosophy it means :

- a. The structure will never break regardless of stress applied
- b. The failed structure can be replaced easily
- c. Adjacent structure withstand the loads which the failed structure should have carried**

167) The best extinguisher to use on a wheel or brake fire is:

- a. CO₂
- b. Water
- c. Dry powder**

168) A constant pressure EDP supplies fluid:

- a. at a variable volume and controls its own pressure**
- b. at a variable pressure and controls its own pressure
- c. at a constant and controls its own volume

169) An Aircraft flying at Mach 1 at low altitude with an air temperature of 15 degree Celsius and an A/c flying at 30000 ft where the temperature is - 43 degree Celsius, the effect on TAS will be:

- a. Lower A/C will have a Lower TAS
- b. Higher A/c will have a lower TAS
- c. Lower A/c will have a higher TAS and Higher A/c will have a lower TAS

170) Which statement is correct:

- a. Flight spoilers rise on the wing with the "up-going" aileron
- b. Flight spoilers remain retracted on the wing with "up-going" aileron
- c. Flight spoiler remain retracted irrespective of the "up-going" or "down-going" ailerons

171) When large aircraft are employed, a hand pump is usually used for:

- a. Ground servicing purposes
- b. Lowering the landing gear in an emergency
- c. Pressuring the oleo struts in the air

172) In fire detection and protection systems, a mechanical system of fire detection which is reversible incorporates:

- a. A thermal switch and copper wire
- b. Carbon fire detection
- c. A small tube filled with helium gas and a titanium wire

173) In a parallel connection/circuit:

- a. It receives the same voltage

b. Failure of resistor will not affect the other

c. All of the above are correct

174) A low bypass turbojet engine has a forward velocity (V) of 600 mph and a jet velocity (V_j) of 1000 mph. Find propulsive efficiency

a. 33%

b. 65%

c. 75%

175) A low bypass turbojet engine has a forward velocity (V) of 200 mph and a jet velocity (V_j) of 1000 mph. Find propulsive efficiency

a. 33%

b. 65%

c. 75%

176) The extreme right limitation a gust load diagram is the speed:

a. V_d

b. V_c

c. V_b

d. V_s

177) What's the limit load factor for a nominal transport category aircraft?

a. 2.5g

b. 1.75g

c. 3.75g

d. 1.5g

178) What is the positive manoeuvre limit load factor for an aircraft in the utility category?

- a. 2.76g
- b. 2.5g
- c. 6.0g
- d. 4.4g

179) ACTION TO BE TAKEN IN CASE OF ENGINE FIRE
correct sequence of operation:

3. PULL No. 1 FIRE HANDLE

1. IF AFTER 30 SECONDS FIRE WARNING REMAINS ILLUMINATED ROTATE No. 1 FIRE HANDLE RIGHT TO ITS MECHANICAL LIMIT AND HOLD FOR AT LEAST 1 SECOND. THIS WILL DISCHARGE THE RIGHT BOTTLE INTO THE LEFT ENGINE.

6. LAND AS SOON AS POSSIBLE

5. CLOSE LEFT THRUST LEVER

4. LEFT ENGINE HP OR ENGINE START LEVER CLOSE

2. NUMBER 1 ENGINE FIRE HANDLE ROTATE LEFT TO MECHANICAL LIMIT AND HOLD FOR AT LEAST 1 SECOND. THIS WILL DISCHARGE THE LEFT BOTTLE INTO THE LEFT ENGINE

- a. 5,4,3,2,1,6
- b. 2,4,6,3,1,5
- c. 6,5,4,3,1,2

180) Generally fuselage design are:

- a. Circular
- b. Oval/ double bubble
- c. All above

181) CB's are

- a. Trip free

b. Trip free/Non trip free

c. Non trip free

182) Generally wind shield consists of -----Layers

a.6

b.4

c.5

183)Choose the correct statement:

a. Control system should be free of backlash

b. Backlash is free movement of cockpit control when the direction of movement is reverted.

c. All of the above

184)Thermal runaway can take place in----- when constant-----

a. Alkaline batteries/current charging system is used

b. Ni/Cad batteries/ current system is used.

c. Alkaline batteries/ Voltage charging system

185)Control locks are used to protect:

a. Mechanical damage in strong gusty wind

b. Controls from wear and tear

c. unauthorized interference with them

186)What kind of sensors are used in IHUMS Systems

a. Accelerometer, tachometers and magnetic pickups

b. Accelerometer, ground speed gauges and magnetic pickups

c. Deformation gauges, tachometers and magnetic pickups

187) What is inertia?

Opposition which a body offers to change in motion.

A large mass has no inertia

Inertia refers to only stationary masses

188) Mark the correct statement:

Vortex generator improves aerodynamics characteristics

Vortex generator increases drag

All of the above

189) One of the factors affecting capacitance is distance between the plates:

A small distance gives a large capacitance

A large distance gives a large capacitance

A small distance gives a small capacitance

190) Circuit breaker normally opens the circuit:

And has the capability of opening and closing the circuit

It may also be used as a circuit isolation switch

All of the above

191) The movement of the flying control surface in response to the movement of cockpit control may be achieved by:

Hydraulically

Electrically

Any one or combination of the above

192) To improve aerodynamic performance, modern aeroplane make use of surfaces that affect the airflow over the surface that could be:

Flaps

Spoilers and speed brakes

All of the above

193) Upto which pressure altitude do most cabin pressure system operates to provide sufficient oxygen supply to the passenger?

35000ft

10000 ft

8000 ft

194) How is the ice prevented from forming on the front wind shield of large systems

Warm air is blown over the wind shield

De-icing fluids are sprayed

Electrically heated wind shields

195) What is the reason for mass balancing a control surface?

To move the CG of the control surface forward and thereby preventing flutter

A form of aerodynamic balance

The stick force will increase

196) Where mass balance weight is located with respect to hinge line:

Always on the hinge line

In front of hinge line

Aft of the hinge line

197) If an aircraft is statically unstable it:

Will only be dynamically stable at high speeds

Will never be dynamically stable

May or may not be dynamically stable, depending on the type of aircraft

198) How does the fin stall occur?

As for any aerofoil there will be critical angle at which fin stalls

Aileron is applied in the same direction to correct the side slip

Both A and B

199) The purpose of diffuser vanes in a centrifugal compressor is to:

Increase the charge temperature

Convert the kinetic energy into pressure energy

Convert pressure energy into kinetic energy

200) The difference between anti icing and de-icing system is:

Anti icing system prevents the formation of ice while deicing system removes the ice after it is formed

De-icing system prevents the formation of ice while anti icing system removes the ice after it is formed

Anti icing is used in air while de-icing is used on ground

201) If the ailerons deflection is increased there will be:

Increment in Angle of bank

Increment in rate of roll

No change in angle of bank

202) A supercharger is capable of maintaining sea level values of power upto a high altitude is called:

Ground boosted supercharger

Externally driven supercharger

Altitude boosted supercharger

203) An automatic cutout valve ACOV will:

Provide idling circuit when selection is made

Extend the life of the accumulator

Provide an idling circuit when the accumulator is fully charged

204) The main advantage of Pneumatic system:

No fire hazard

Lighter system because no return lines are required

All of the above

205) Inadvertent retraction of the landing gear on ground is :

Not possible as the system is not powerful enough

Prevented by the ground/air logic system

A danger after the aircraft is started

206) In an emergency decent How can the rate of decent are increased?

Reduce thrust by closing throttle

Extend speed brakes, Increase speed Mmo or Vmo or Vle if the gear is down

Both A & B correct

207) Wing flaps extended speed (Vfe) is :

The minimum air speed at which the aircraft should be flown

The minimum air speed at which the flaps can be extended and the aircraft can be flown

The maximum air speed at which the aircraft should be flown with flaps in prescribed extended position

208) Choose the correct : As an aircraft climbs out of ground effect the following changes take place

The increase in downwash will produce a nose up pitching moment

The nose up pitching moment may induce an inadvertent over rotation and tail strike

Both A & B

209) APU are self monitoring and will shut down in the event of :

Fire and oil pressure

Fire, Oil pressure failure, Overspeed and Overheat

Fire and overheat only

210) With asymmetry thrust mark the correct:

Effectiveness of controls depends upon the static pressure

Lower IAS will reduce the available rudder movement

Both A & B

211) CP moves:

FWD at increased transonic speed

Remain constant in sweep back wing

Aft at increased transonic speed

212) Icing on aircraft:

Can contaminate wing and tail surface, engine and pitot static system

Can change the stall characteristics and stall warning devices may not function

Both A & B

213) Chemical oxygen generators:

The generator are relatively light self contained devices

Sufficient oxygen is supplied for a period of minimum 30 minutes

A relief valve prevent internal pressure in the generator exceeding 50psi as normal flow rate is 10 psi

214) An atom that has lost an electron becomes:

More negative

More positive

Static

215) The magnetic chip detectors are fitted in:

- a. the pressure line between the pressure pump and the engine.
- b. suction line between the reservoir and the pressure pump.
- c. return line between the engine and the scavenge pump.**
- d. return line after the FCOC.

216) Gas turbines use for lubrication:

- a. mineral oil with additives (compound).
- b. mineral oil straight.
- c. multi-grade 20/50.
- d. synthetic oil.

217) Effects of an electric current: (READ)

Heating Effect. When a current flows through a conductor it always causes the conductor to become hot - electric fires, irons, light bulbs and fuses.

Magnetic Effect. A magnetic field is always produced around the conductor when a current flows through it - motors, generators and transformers.

Chemical Effect. When a current flows through certain liquids (electrolytes) a chemical change occurs in the liquid and any metals immersed in it - battery charging and electroplating.

218) Downwash :

Behind the leading edge

Downwash behind trailing edge

Upwash behind the leading edge

219). Component along plane of rotation

Torque

Thrust

220) Hydraulic accumulator

Dampen pressure fluctuations

Back up due to pressure pump failure

Both a and b

221) Integration of aerodynamic load and elastic deformation

Aero elasticity and aeroelastic coupling

Fluttering

222) High energy ignition unit ignites

4 times in one revolution

60-100 times in one second

60-100 times in one minute

223) Control surface flutter occurs because of

Unbalance

Overbalance

Mass balancing

REDBIRD
FLIGHT TRAINING

224) The alpha range of a variable pitch propeller is between:

- a. feather and flight fine pitch stop**
- b. feather and ground fine pitch stop
- c. flight fine pitch stop and reverse stop
- d. ground fine pitch stop and reverse stop

225). Coring

Oil temperature increase

Oil pressure increase

Oil temperature increase then decrease

(A problem that can occur during starting in very cold weather is Coring. It is caused by the fact the cold viscous oil does not flow correctly through the engine.)

(To overcome coring oil cooler flaps should be Closed, this will initially increase temperature but should improve flow particularly through the cooler, and then bring temperatures down.)

- 226) On a part throttled engine, take-off thrust would be achieved:
- a. later than normal due to pressure in the compressor being low
 - b. later than normal due to the EPR being low
 - c. at less than full throttle position**
 - d. later than normal due to the EPR being high

227) Turbine converts

Pressure to kinetic energy

Kinetic to pressure energy

Kinetic velocity to pressure velocity

228) The power output of a piston engine can be calculated by multiplying:

- a. force by distance
- b. work by velocity
- c. pressure by moment arm
- d. torque by rpm**

229) Wing underside stress in level flight, no turbulence

Tension

Shear

Compression

230) Short wing span

Increase in rate of roll as compared to large wing span

Decrease in rate of roll as compared to large wing span

No change in rate of roll

231) How many speeds does aircraft is in equilibrium with required power by available in power.

232) Effects of temperature in battery

Temperature affects battery

Low temperature decreases rate of discharge

Lead acid has improved performance on high temperature

(ALL THREE ANSWERS ARE CORRECT)

233) for battery charging

Constant voltage

Constant current

None of the above

234) Limit load factors are based on the

Mean weight of aircraft

Max weight of aircraft

Max weight divided by lift